

**Knowledge Graphs and Information Retrieval Systems**  
**D17A/B/C**

| Exp. No. | Experiment Title                                           | Student Roll Numbers | Brief Explanation                                                                                         | Expected Outcome                                                                        |
|----------|------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 1        | Document Processing Pipeline for Information Retrieval     | 1-5                  | Build a complete pipeline to acquire, clean, tokenize and prepare textual documents for retrieval.        | Students will understand the end-to-end lifecycle of documents in an IR system.         |
| 2        | Text Preprocessing and Normalization                       | 6-10                 | Perform tokenization, stop-word removal, stemming, lemmatization and text normalization.                  | Clean and standardized text corpus ready for indexing and analysis.                     |
| 3        | Construction of an Inverted Index                          | 11-15                | Create an inverted index mapping terms to documents and their occurrences.                                | A basic searchable index supporting efficient keyword retrieval.                        |
| 4        | TF-IDF Based Document Retrieval                            | 16-20                | Represent documents and queries using TF-IDF and rank documents based on similarity.                      | Students will implement a traditional vector-space retrieval system.                    |
| 5        | BM25 Based Document Ranking                                | 21-25                | Implement BM25 and compare its ranking performance with TF-IDF.                                           |                                                                                         |
| 6        | Identify Graph Entities                                    | 26-30                | Identify entities such as persons, organizations, locations and other domain-specific concepts from text. | Understanding of probabilistic ranking and comparative retrieval performance.           |
| 7        | Relationship Extraction from Text                          | 31-35                | Extract semantic relationships between identified entities.                                               | Entity-relationship triples suitable for graph construction.                            |
| 8        | Create and Manage a Graph Database                         | 36-40                | Install Neo4j, create nodes and relationships, and perform basic graph operations.                        | A functioning graph database containing connected entities.                             |
| 9        | Design a Knowledge Graph Schema and Import Data            | 41-45                | Design node labels, relationship types and properties, then import structured entity-relationship data.   | A domain-specific knowledge graph implemented in Neo4j.                                 |
| 10       | Query Knowledge Graphs using Cypher                        | 46-50                | Write Cypher queries to retrieve nodes, relationships and multi-hop connections.                          | Ability to retrieve meaningful information from knowledge graphs.                       |
| 11       | Advanced Cypher Queries and Graph Pattern Matching         | 51-55                | Perform multi-hop traversal, filtering, aggregation and complex graph pattern queries.                    | Efficient retrieval of complex relationships and hidden connections.                    |
| 12       | Dense Embedding-Based Semantic Search                      | 56-60                | Generate dense vector representations of documents and queries and perform similarity-based retrieval.    | A semantic search system capable of retrieving conceptually similar documents.          |
| 13       | Hybrid Keyword and Semantic Retrieval                      | 61-65                | Combine lexical retrieval (BM25) with embedding-based semantic search.                                    | Improved retrieval performance through complementary retrieval strategies.              |
| 14       | Integration of Information Retrieval with Knowledge Graphs | 65-70                | Combine document retrieval results with graph-based entity and relationship exploration.                  | A graph-enhanced information retrieval system that supports contextual exploration.     |
| 15       | Evaluation of Retrieval Systems                            | 71 onwards           | Evaluate different retrieval approaches using Precision, Recall, F1-score, MRR and related metrics.       | Quantitative comparison of keyword, semantic, hybrid and graph-based retrieval systems. |